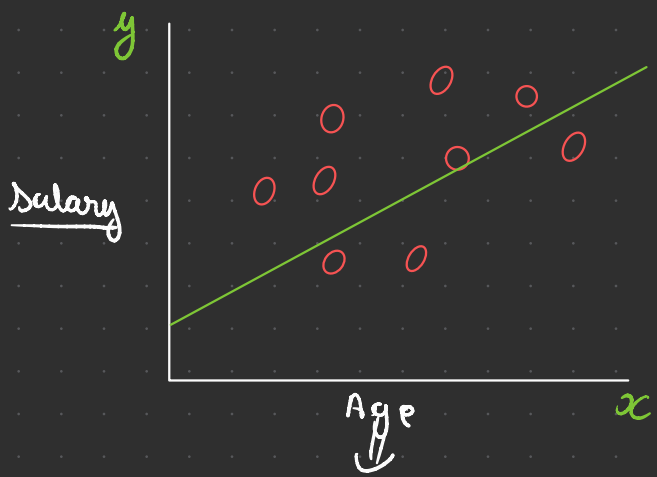
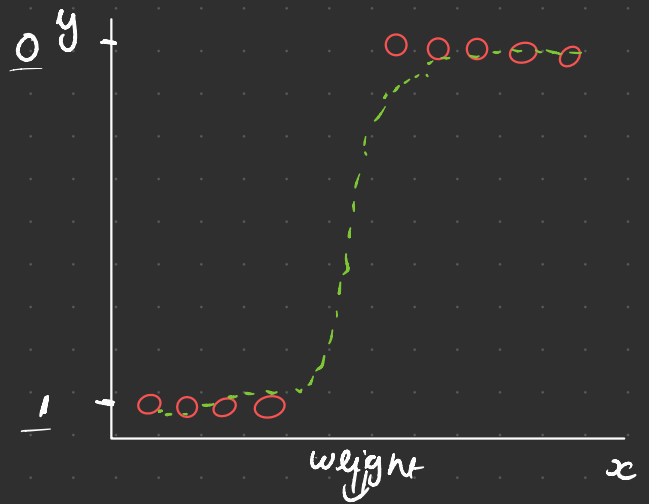


Logistic regression

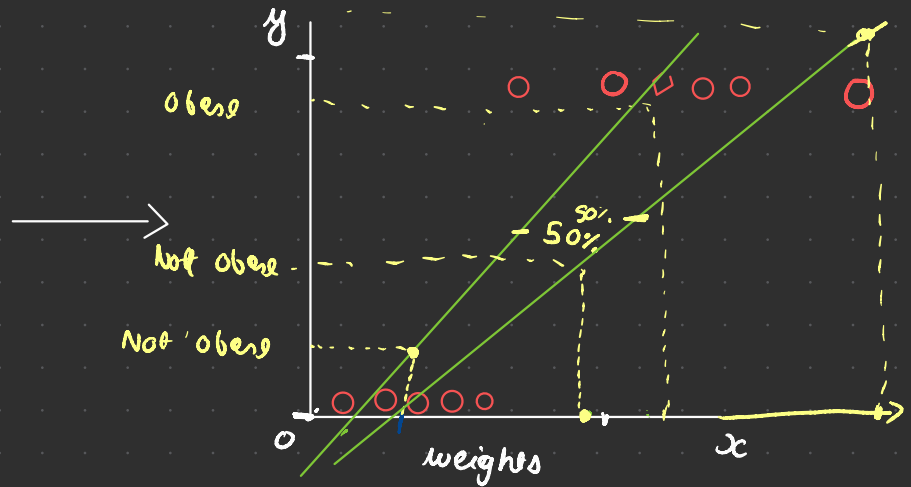


Linear Regression



Logistic Regression

weight	Obesity
92	yes
63	no
24	no
102	yes
196	yes
268	:



To create the best fit line

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1$$

① Outliers → Influence the straight line

② Result can be negative < 0 and Positive > 1

So to prevent all of these things ↗
we squash the line

to make the straight into S shaped curve we will use **SIGMOID** activation function



$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1)$$

$$g = \frac{1}{1+e^{-z}} \rightarrow \theta_0 + \theta_1 x_1$$

$$h_{\theta}(x) = \frac{1}{1+e^{-(\theta_0 + \theta_1 x_1)}}$$

$$h_{\theta}(x) = \frac{1}{1+e^{-(\theta_0 + \theta_1 x_1)}}$$

$e \rightarrow$ eulers number $\rightarrow$ A mathematical constant
 $\Rightarrow 2.71828$.

$e \rightarrow$ gives smooth curve

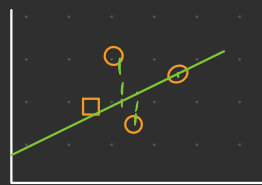


Linear Regression

$\theta_0, \theta_1 \rightarrow$ values $\rightarrow$ to create best fit line

Cost function

$\hookrightarrow$



error

Logistic Regression

$\theta_0, \theta_1 \rightarrow$ values $\rightarrow$ to create best Squashed line

Log loss function

$\hookrightarrow$ loss

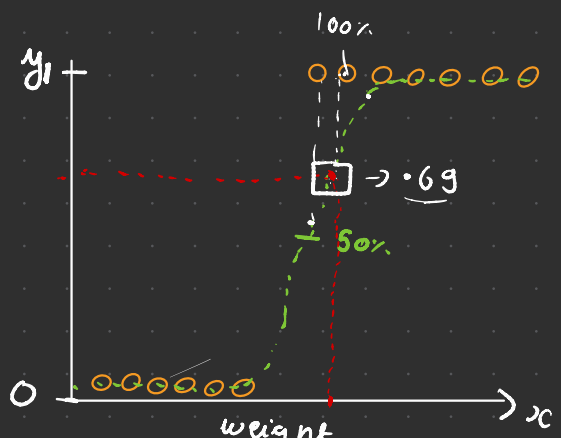
Log loss (Binary cross entropy)

$(0,1)$

lets say Predicted Probab $\Rightarrow 0.99$ and actual is 1

$\Rightarrow$ very less loss

obesity



Let say my predicted probability = 0.01 and actual is 1

⇒ very high loss → Penalty — is calculate with log loss function

$$\text{log loss} = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)}) \right]$$

m = number of sample

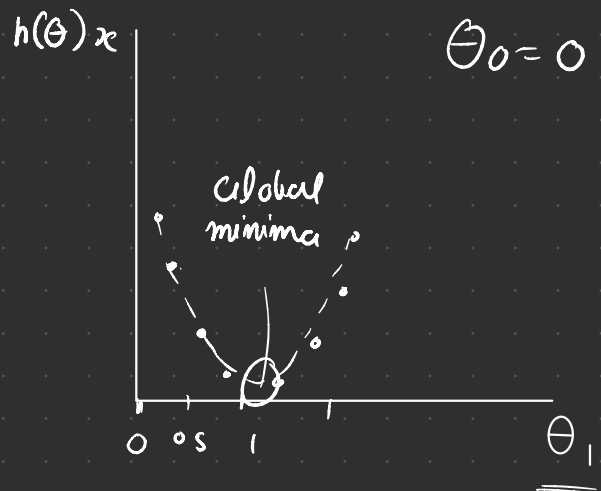
y^i = actual label (0, 1)

$\hat{y}^i$ = Predicted Probability (0 - 1)

$$y^i = 1$$

$$\hat{y}^i = 0.54$$

$$\text{Log loss} = 0.616$$



$\theta_0, \theta_1 \rightarrow$ Perpet values

Evaluation of model

250

Confusion matrix

		Predicted		
		1	0	
Actual	1	TP	FN	Type 1 error
	0	FP	FP	

190	10
10	40

Type 2 error

- ① Accuracy
- ② Precision
- ③ Recall
- ④ F1 Score

① Accuracy

→ The percentage of correct prediction

$$\Rightarrow \frac{TP + FP}{TP + FP + FN + FP} = \frac{190 + 40}{190 + 10 + 10 + 40} = 0.92$$

92 %

② Precision

Out of all predicted positives, how many were actually positive

$$\Rightarrow \frac{TP}{TP + FP} = \frac{190}{190 + 10} \Rightarrow 0.95 \%$$

Use when false positive are costly (spam detection)

③ Recall

Out of all actual positive, how many did we correctly predicted

$$\Rightarrow \frac{TP}{TP + FN} = \frac{190}{190 + 10} = 0.95 \%$$

Use when false N are costly (disease detection)

④ F₁-score

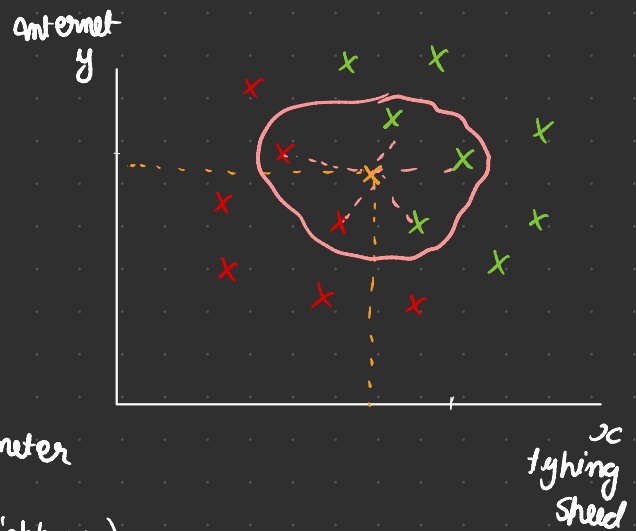
⇒ harmonic mean of Precision and Recall -

$$F1\text{ score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Use when you want balance b/w Precision & Recall

K NN → K Nearest Neighbour

→ Internet	→ typing speed	hacker
<u>24</u>	<u>36</u>	1
<u>12</u>	<u>22</u>	0
⋮	⋮	⋮



- ① Define K value → $K=5$ - hyperparameter
(5 nearest neighbour)
- ② Calculate the Distance from all points (euclidean Distance)
- ③ ↳ Sort all the Distance → 3 yes, 2 no
- ④ Majority count →
↳ yes the person can become a hacker

Other than classification, → mean of 5 nearest neighbour

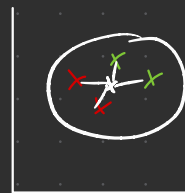
you can use n dimensions



$k \rightarrow$ value must be an odd number

experiment

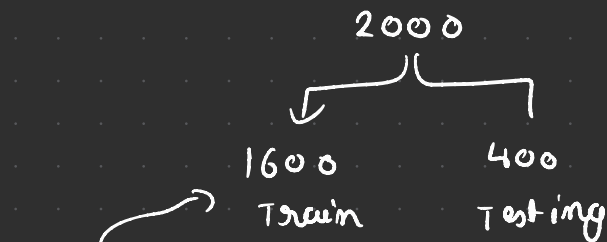
$\hookrightarrow$ cross validation



$k = 4$

yes - 2

no - 2



$$k = 5 = 86\%$$

$$k = 3 = 89\%$$

$$k = 1 = 60\%$$

$$k = 7 =$$

$\vdots$

$$k = 25 = 92\% \quad \checkmark \uparrow \uparrow$$

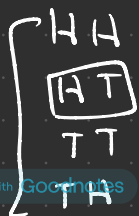
Naive bayes

Independent

$$\text{Coin} = P(\underline{H}) = \frac{1}{2}, P(\underline{T}) = \frac{1}{2}$$

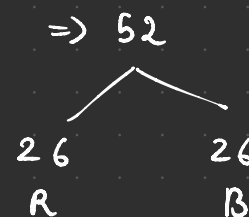
$$\text{Dice} = P(6) = \frac{1}{6}$$
$$P(4) = \frac{1}{6}$$

$$P(A \text{ and } B) = P(A) \times P(B) \quad \text{--- (1)}$$
$$= \frac{1}{2} \times \frac{1}{2} = \boxed{\frac{1}{4}}$$



Dependent

Deck of cards



$$P(R) = \frac{26}{52} = \frac{1}{2}$$

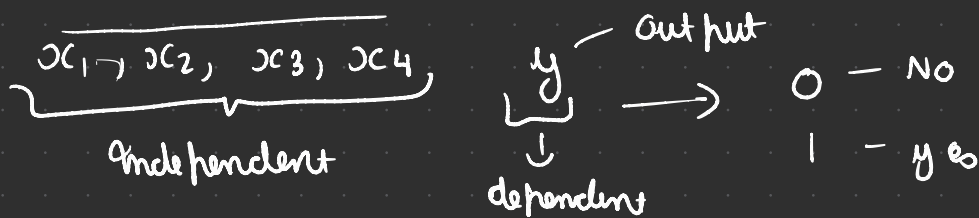
$$P(R) = \frac{25}{51} = ?$$

$$P(A \text{ and } B) = P(A) \times P(B|A) \quad \text{--- Conditional Probability}$$

$$P(A \text{ and } B) = P(B \text{ and } A)$$

$$P(A) \times P(B|A) = P(B) \times \underline{P(A|B)}$$

$$\boxed{P(A|B) = \frac{P(A) \times P(B|A)}{P(B)}} \rightarrow \text{Bayes Theorem}$$



$$\underline{P(A|B) = \frac{P(A) \times P(B|A)}{P(B)}}$$

$$P(y | x_1, x_2, x_3, x_4) = \frac{P(y) \times P(x_1, x_2, x_3, x_4 | y)}{P(x_1, x_2, x_3, x_4)}$$

↓

$$\text{Yes} = P(y | x_1, x_2, x_3, x_4) = \frac{P(y) \times P(x_1 | y) \times P(x_2 | y) \times P(x_3 | y) \times P(x_n | y)}{P(x_1) \times P(x_2) \times P(x_3) \dots P(x_n)}$$

$$\text{No} = P(N | x_1, x_2, x_3, x_4) = \frac{P(N) \times P(x_1 | N) \times P(x_2 | N) \dots P(x_n | N)}{P(x_1) \times P(x_2) \dots P(x_n)}$$

Decision Tree

Tree $\rightarrow$ Branches, Leaves, Root

money = ?

if money < 30 :

Snack

elif money > 30 and < 60 :

fast food

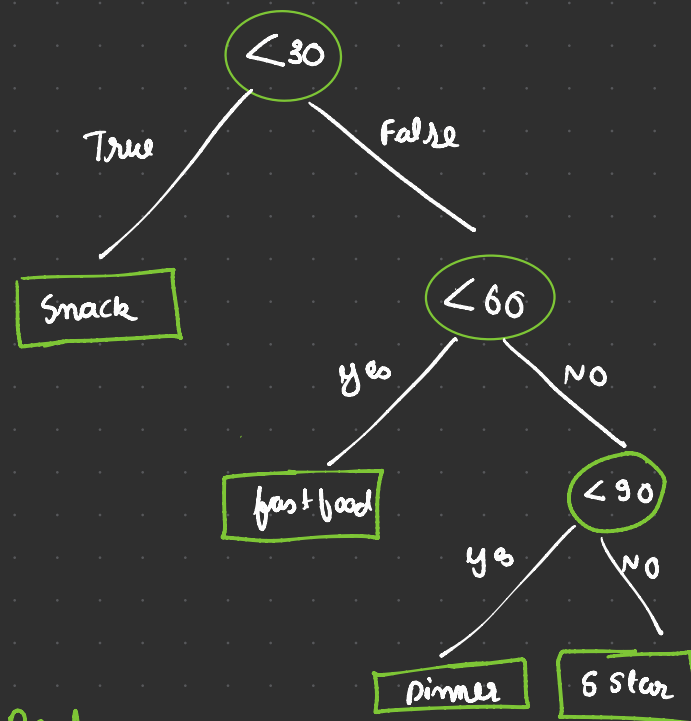
elif money > 60 < 90 :

Dinner

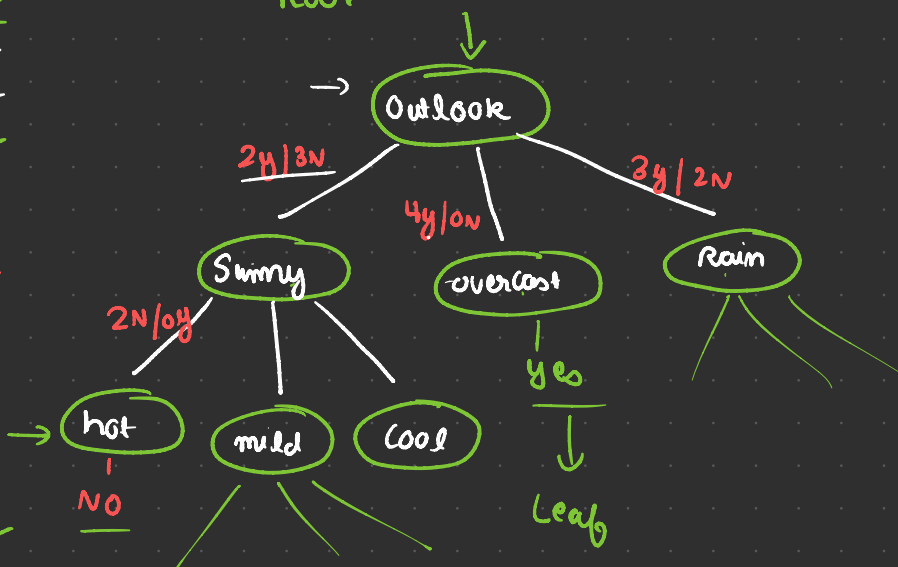
else :

5 star

Outlook	Temperature	Humidity	Wind	Play
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No



Root



Entropy, Information Gain

for choosing Root node

- ① → Start with the Data
- ② → choose the best feature
- ③ → make branches
- ④ → Repeat the process for each branch
- ⑤ Stop when :

All data is here
max depth is Reached

Outlook	Temperature	Humidity	Wind	Play
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

$$\text{Entropy}(S) = -P_+ \log_2(P_+) - P_- \log_2(P_-)$$

$$\text{Play} = 5 \text{ No}, 9 \text{ Yes}$$

$$= -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right)$$

$$\Rightarrow 0.940$$

Outlook

↳ Sunny → 2 Yes 1 No

↳ Overcast ⇒ 0

↳ Rain ⇒ 0.971 S

$$= -\frac{2}{5} \log_2\left(\frac{2}{5}\right) - \frac{3}{5} \log_2\left(\frac{3}{5}\right) = 0.971$$

Information Gain

$$IG(S, \text{Outlook}) = \text{Entropy}(S) - \sum_{\text{values}} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

$$0.940 - \left[\frac{5}{14} \times 0.971 + \frac{4}{14} \times 0 + \frac{5}{14} \times 0.971 \right]$$

outlook $\rightarrow 0.246 \Rightarrow$ Information Gain $\star$

Temp $\rightarrow \sim 0.029$

humidity $\rightarrow \sim 0.191$

wind $\rightarrow \sim 0.048$

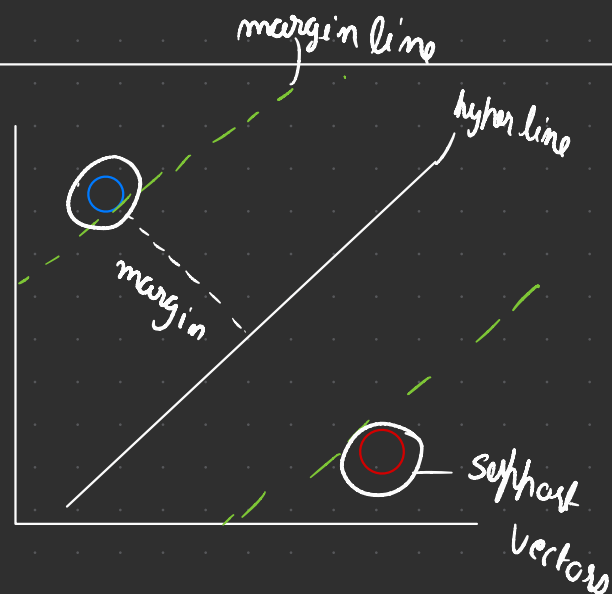
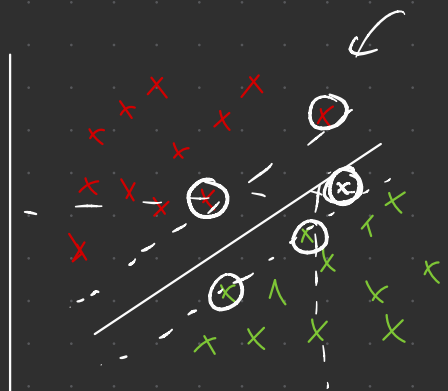
SVM (Support vector machine)

Red ball

blue ball

2D = hyperline

3D = hyperplane



$$\Rightarrow wx + b = 0$$

x = point you want to predict

w = weight / slope

b = bias

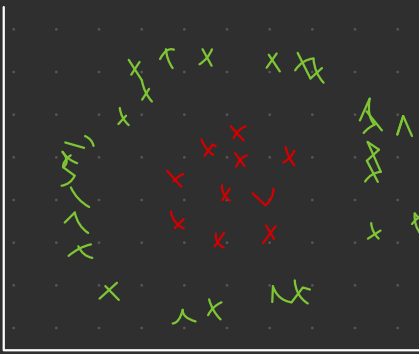
$$wx + b > 0 \rightarrow \text{class 1 - Red}$$

$$wx + b < 0 \rightarrow \text{class 2 - Green}$$

hyperplane $\rightarrow$ A line that separate two categories

Margin $\rightarrow$ the space b/w the line and nearest point

Support vector $\rightarrow$ the closest ball to margin that support the line



→ we use kernels to distinguish b/w
Red and Green Points